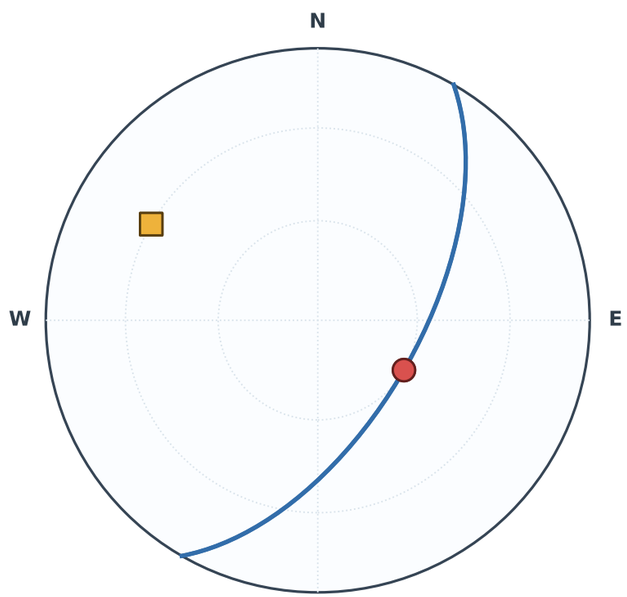
Geographic rotation: eight routes to the same fabric

All stereonets are lower hemisphere, north up and east right. Each blue trace and symbol overlays eight plotting conventions and specimen-axis choices.

A Field case from CSV

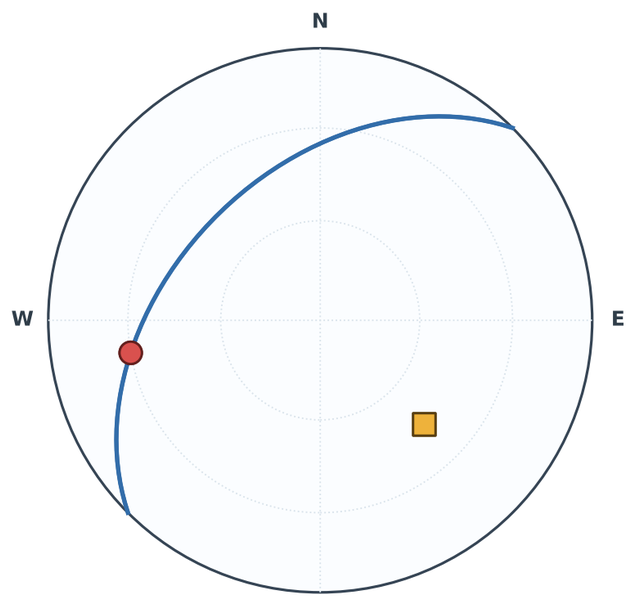
SYNTHETIC DOWN-DIP · strike 30°, dip 60° SE · trend 120.0°, plunge 60.0°



8 overlays · greatest plotted-coordinate difference 2.2e-16

B Synthetic oblique grid case

Synthetic oblique grid case · strike 225°, dip 45° NW · trend 260.3°, plunge 30.0°



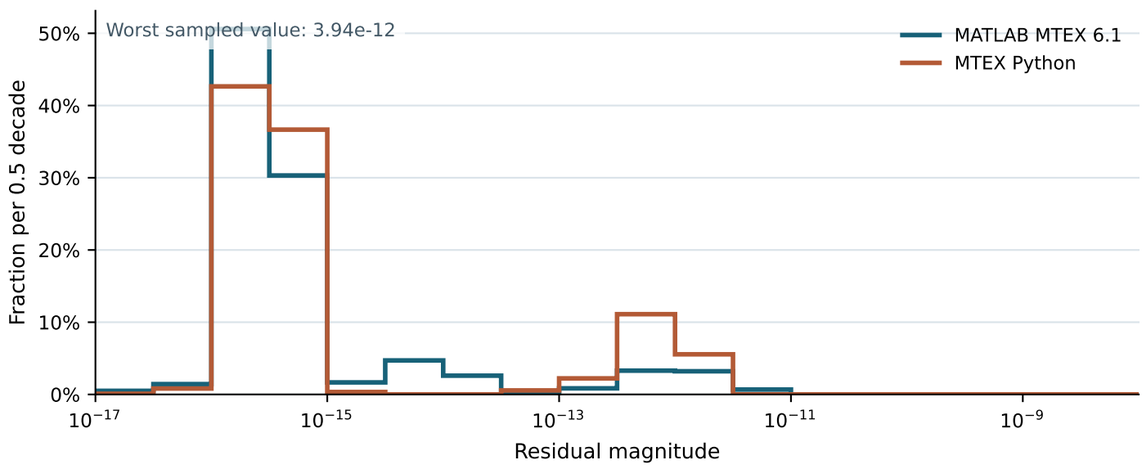
8 overlays · greatest plotted-coordinate difference 2.2e-16

Blue line: foliation great circle ■ downward foliation pole ● lineation (on the plane)

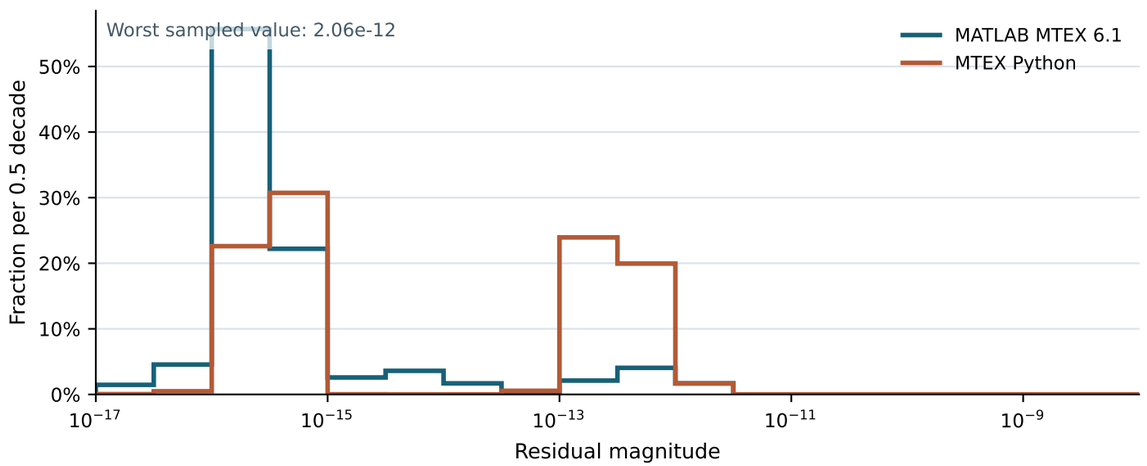
C Residual density across the full matrix audits

MATLAB: 568,320 mappings · Python: 466,560 mappings · separate deterministic grids, each spanning eight conventions

Mapped line / pole: unit-vector error



Geographic equal-area XY error



Each distribution contains every mapping's larger line/pole residual. XY coordinates use a unit-radius equal-area horizon in both audits. Values ≤10^{-17} share the first bin.

Default CSV case is synthetic; all specimen axes are illustrative. Exactly vertical oblique trend-only cases are underdetermined and excluded from the grid.